Building a Unified Science of Natural Behavior with Densely Overlapping Measurement Environments [Domes]

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

1. Mission

> Develop and disseminate a new class of scientific instrument called a **Densely Overlapping Measurement Environment (Dome)** that records the complete perception–action loop of a behaving agent as a single calibrated, uncertainty-tagged measurement, and build the autonomous organization dedicated to the measurement itself and form the boundary object that will align perceptual and motor neuroscience, musculoskeletal biomechanics, mobile robotics, and embodied AI into a single convergent science of sensorimotor control.

1.1. A New Observable: The Sensorimotor Loop

Every agent, living or engineered, must solve the same problem - sense a thin slice of available energy in the environment through a limited set of imperfect transducers and on its basis push its body against the world to propel itself towards its goal. Information flows in, forces flow out; the brain exists to yank the bones around.

We can measure individual components of this loop with extraordinary precision progressing at the rate of our technology. Camera and IMU based motion capture can track the kinematics of the body, force plates measure the kinetic forces between the body and world. Outward facing cameras can measure the world, and inward facing cameras can track the eyes. For humans, EEG nets and surface EMG record course records of patterns of neural activity in the cortex and musculoskeletal system. In animal models, dense electrode arrays resolve the individual spikes of neurons and motor units.

And yet, despite the extradradoinary precision and progress along these individual threads of measurement, access to the coherent whole representing the shared observational context that each of these tools illuminates remains ellusive. Occasionally, a research team will reach accross disciplinary divides and stitch several threads together for heroic integrative studies, but the methods rarely escape the Methods sections of a small cluster of publications spanning roughly the timescale of a single PhD.

Each of these teams sought out to answer a question at the edge of their discipline and instead found themselves in the business of tool building. The build these tools because they know the answers to the questions that they seek lie in the output of the integrated tool that does not exist. In an institution that rewards the plating of seeds over the tending of gardens, the long and laborious work of maintaining a novel complex instrument quickly overwhelms the capacity of an academic lab. Eventually the nascent tool collapses under the weight of its own complexity, and what was meant to be a bridge between two domains of study instead fades into an evergrowing pile of academic abandonware.

We need a new organization that exists outside of the siloed domains of the Academia that dedicates itself to the measurement, the development and dissertation of world-class, convival tools and novel skillsets explore the landscape the capacity unlocks. That organization is the **FreeMoCap X-Lab** [FMC-X]. We have already changed the landscape human-focused research by bringing high-quality capture to over 15,000 researchers, clinicians, robotists, and animators across 152 countries. With the support of the NSF X-Labs program, we will extend our mission and change entire landscape of neuroscience, biomechanics and robotics.

1.2. A New Instrument: Building the Dome

We define a Dome as a densely instrumented region of real space engineered (at some aspirational extreme) to record every possible measurable channel of the agent–environment interaction at once. It is the opposite of a space telescope, an empty volume of of lenses focused inward at the mysterious place where the body meets the world.

In practice, individual Domes will comprise a partial subset of available instrumentation, selected based on use-case. However, criticall, **All Domes produce the same output** regardless of the component instruments - a metrologically-grounded record of semantically-coherent cannonical models **partially hydrated** by measurements from the available instruments [TODO - Define/invoke Sensor-Grounded Ontology thing here].

A motion capture recording produces the same reconstructed `Head` A mocap `Skull` is assumed to have `Eyes`, reagardless of whether the subject was wearing an eye tracker to hydrate its Degrees of Freedom. Both a ferret and a human have a `Skull` wqhich maybe approxomated as the same `RigidBody` defining segments of the body of a humanoid robot (SH, CH, JH, GN. The reconstructed `RetinalInput` derived from reconstructions of the `Body`, `Eye`, and `Environment` of a human participant running across a rocky field MH/KB are precisely the same data models as those derived from a `Marmoset` leaping from branch to branch with Neuropixel electrodes emdedded in its `SuperiorColliculus` leaping to platform AH/JY, a `Ferret` chasing a fictive prey with a Miniscope viewer mounted over of its `PosteriorParietalCortex` BS, and a `GuineaFowl` running across a pneumatically actuated platform with EMG-electrodes implanted in its `MedialGastrocnemius` MD.

[TODO - reference specific work building DOME-variants in flagship facility and fan-out of built-to-use variants to collaborator network. Mention targetting specific hardware based on practical and research-grounded needs - target Retinal input good enough to target ephys, needs build better eye tracker (dense sensor array for better eye data in general, and adding unmeasurable DoFs like torsion and accomdation for accurate retinal optic flow(CURL). Need world sensor to map environment, built into head-mounted world sensor array of eye tracker, later expanding to attach that scanner to drones. Auto-calibrating camera sensor array - break practicality plateus of large capture volumes, allow for autonomously driven experiments on ideal camera configureations, and ai-driven autonoumous collaborative experiments by intergrating drone-swarms in the scene (i.e. the drone swarms and camera arrays interact to test different capture configurations, training the drone-swarm control policies, all which preps for integrations into Dome-Mobile later to ground inside-out sensors of IMU-suit and head-mounted world scanner))]

1.3. The New Capability

Because every channel is spatially calibrated, temporally synchronized, uncertainty-traced within a unified sensor-grounded ontology, Dome records drop into modern reinforcement-learning and robotics stacks with minimal reshaping — giving embodied AI the real-world, uncertainty-bounded sensorimotor corpora that safe learning requires and that no current dataset provides.

Internally, each sensor stream is modelable agaist every other stream, both within recordings (modelling eye movements against head movent to model VOR) and across recordings (back-fill old datasets with models of Cyclotorsion after we build a new eye tracker that measures torsion). The data also provides internal automatic erorr signals - downstream estimates of body kinematics fused from IMU- and Camera-mocap can back project onto the the original cmaera images, defining a reprojection-error signal that can improve image pose-detection algorithms that are currently stuck on an increasingly long-in-the-tooth [TODO - synonym] COCO dataset.

The same **Sensor-Grounded Ontology** that [...allows us to align outputs for heterogoenous Domes...] makes the instrument legible to a human — explicit entities, explicit relationships, no assumed domain knowledge — makes its records natively legible to a machine: typed, grounded data over which a model can build reliable reasoning chains rather than guessing at unlabeled streams. This cleanliness is not a layer added on top; it is the property that lets a Dome route its live streams and logs through a local model to assist novices, experts, and developers alike, and that lets published results tag back into one shared entity–relationship structure connecting them to the wider body of science.

1.4. Unmet Needs

An instrument like this is unmet by existing organizational structures and funding mechanisms for reasons that are structural, not incidental. Building a scientific instrument is **a distinct craft, not a downstream application of domain expertise** — mastery of electrophysiology or biomechanics does not transfer to the engineering of a calibrated, maintainable measurement system, and because that difficulty is invisible from outside the craft, integrated measurement is repeatedly scoped as a student side-project, under-resourced, and abandoned when it proves to be a career of engineering rather than a semester of it. The deeper failure is not that these efforts collapse but that **even their successes do not compound**: a research result and a research instrument are different deliverables, and the second — a one-off tuned to a single rig, undocumented and unmaintained — rarely survives the trainee who built it. The field rewards the demonstration, not the durable artifact, and often cannot tell them apart, so proofs of concept are celebrated as finished tools and the unglamorous engineering that would turn one into an instrument others can trust is never done.

Where durable scientific instruments do exist, they were built by escaping this structure rather than succeeding within it — in institutes engineered for the purpose, or by rare leaders who treated maintenance as first-class against their own career incentives. That escape is the necessary condition. By Conway's Law, an enterprise of domain-named departments, journals, and grants can only produce domain-scoped instruments, reviewed by people who score work inside a single domain — never the long metrology that turns a proof of concept into shared infrastructure. A tool defined by a research domain partitions its users into specialists; a tool defined by a measurement unites everyone who needs that measurement and becomes a commons precisely because it imposes none of their goals. The FreeMoCap Foundation already ran this experiment: scoped to the measurement and refusing to claim a domain, it became a commons of over 15,000 users across 152 countries — but a commons cannot form around abandonware, so **the scoping argument and the engineering argument are the same argument**.

The measurement therefore needs its own organization. The PI left a tenure-track position because the institution could not support this work, and founded the FreeMoCap Foundation as a 501(c)(3) whose primary output is open scientific instruments; the X-Lab's operating model follows directly — scope the deliverable to the instrument, hold full-time career engineers for years so the instrument accumulates the mastery that makes it trustworthy, and fan the science out to a standing network of collaborators who use the tools and feed insight back. This is not a critique offered from the sidelines. We know this failure mode because **we built the thing that does not exhibit it**: a working animal-scale instrument and a global user community, engineered to be installed, run, and extended by people who did not build it — exactly the standard the X-Labs model exists to fund, and one a university appointment cannot sustain.

1.5. The Vision

The vision of this proposal represents an grand acceleration of movement the FreeMoCap Project has fostered since 2021, and there is no upper limit to our ambition. The global spread of the FreeMoCap software shows that the world has a hunger for this kind of tool, and we intend to feed it. We will put a Dome in every research lab, every classroom, every Physical Therapy clinic, animation studio, and athletic facility. We will create field-deployable wearble Dome-Mobile systems that provide lab-quality data of humans and animals in the remote wilderness. We will build them into the SCUBA systems of deep sea welders and into the EVA suits and HAB volumes of the ISS. Every country hospital and retirement home will have an AI-enabled, auto-calibrating, self-analyzing Gait and Posture facility that will have plain language conversations with patients about their own kinematic data. We will build Domes into the sides of moutains to measure the aerodyanmics of hunting hawks, and onto the hulls of shipping boats to understand the mechanics of the dolphins that play in their wake. Every Zoo enclosure will become a 24/7 data stream, with animal tracking models jumpstarted through citizen science and then accelerated through auto-training backprojection pipelines. Previously disconnect specialized results on animal models and niche domains will connect through shared links in a Sensor Grounded Ontology, rather than being scattered across publications in specialized journals. The resulting web of scientific beliefs about neuroscience and biomechanics will trace backwards a shared ontology underpinning a unified science of natural behavior fed by every calibrated Dome, be it a $1,000,000 flagship facility or $100 student kit. so that ever reconstructed behavior embodies the same, eternal question - Would you like to know more?

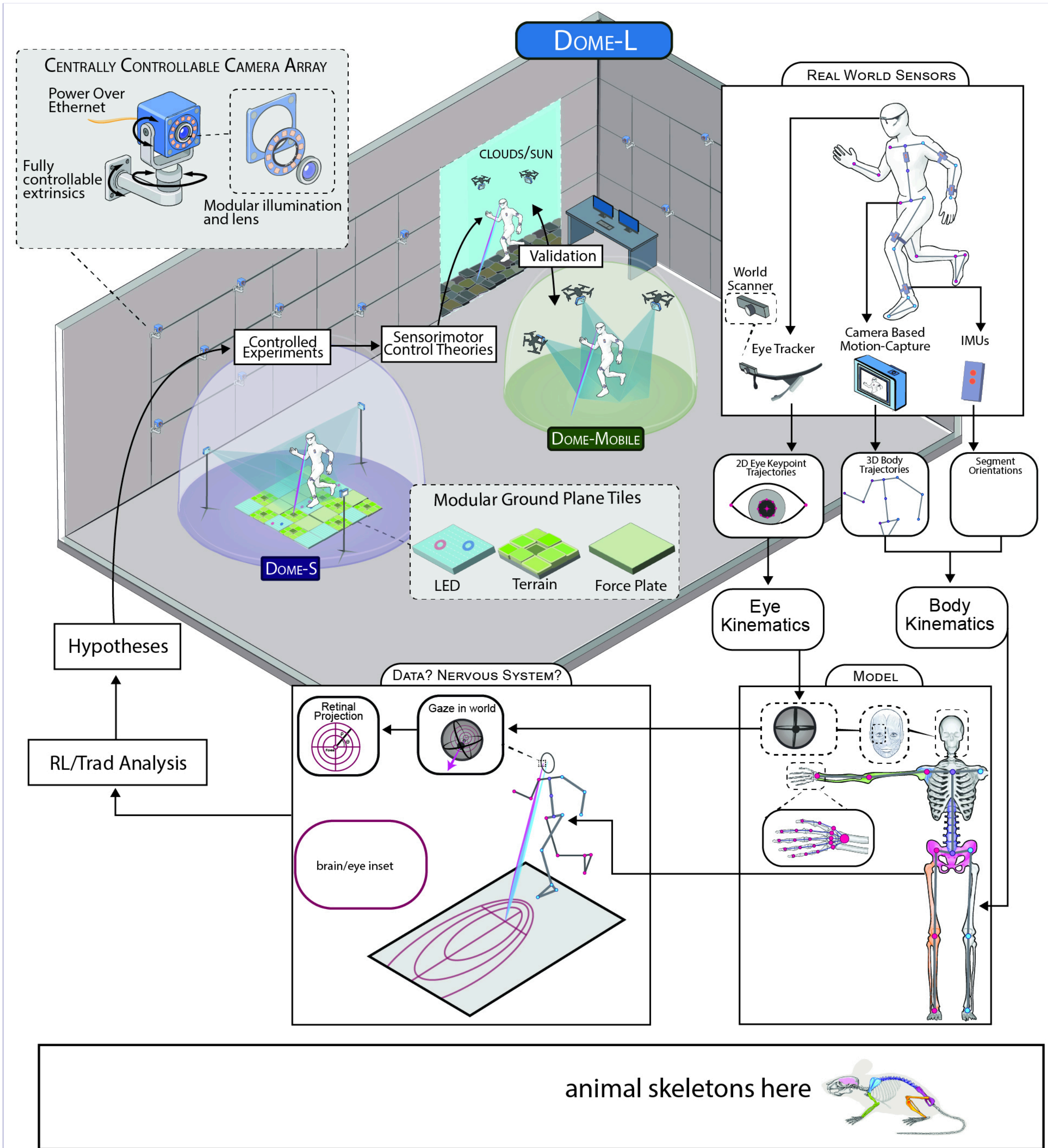

Fig. 1: Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions.

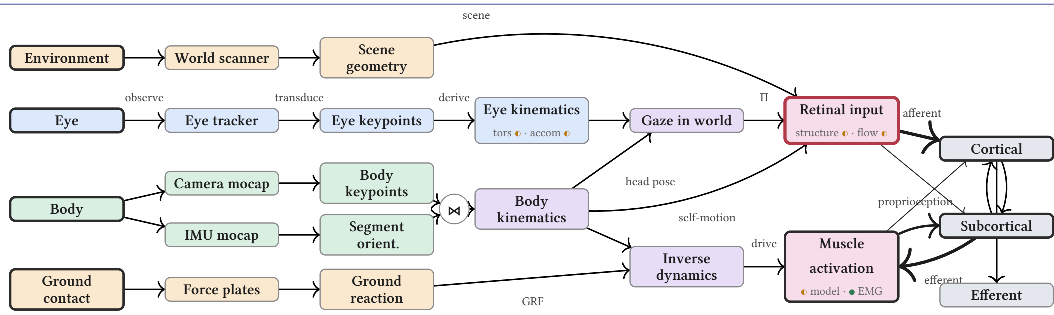

Fig. 2: **The Dome measurement chain.** Each sensor transduces a true fact into signal; the chain derives, fuses, and composes estimates whose channels are ● measured or ◐ inferred under a prior, and projects (Π) them onto the **retinal input** and **muscle activation** that drive the nervous system. The metrological target: drive retinal-input error low enough to **predict neural activity in marmoset visual cortex / superior colliculus** — a ground-truth test no existing system can pose.

Item	In place	Phase 0 · 9–12 mo	Phase 1 · 24–36 mo	Phase 2 · variable
ORGANIZATION — OPERATIONAL DELIVERABLES				
Legal & governance	✓ 501(c)(3)	Governance + Autonomy Plan	board seated	
Facility (lease)		lease + move-in	office fit-out	
Full-time team	✓ core FT	hire · onboard	full staff	
Required NSF plans		Data · IP · Security · Gov; Ph1 plan + budget		
IP management	✓ prior IP	IP Mgmt Plan	license · manage	
BODY — TECHNICAL MATURITY				
Camera array		PoC	Prototype	Operational
IMU↔camera fusion	✓ FMC mocap	PoC	Prototype	Validated
EYE — TECHNICAL MATURITY				
Eye tracker (torsion)		Breadboard	Prototype	Operational
WORLD — TECHNICAL MATURITY				
World-scanner		Concept	Prototype	
Floor panels (force/LED)		PoC	Prototype	
ARGPv3	✓ ARGPv1		Prototype	Validated
Drone swarm			Concept	PoC
INTEGRATION — FIRST-ORDER				
Dome-L (Auto)		PoC / testbed	Prototype	Operational
Dome-S (Static)	✓ FMC webcam	Breadboard	Validated	Operational
Dome-Mobile		Concept	Prototype	Validated
Dome-Animals	✓ ferret (6) · mouse (3)	marmoset	guinea fowl	rat

Table 1: Maturity ladder (TRL): Concept 2 · PoC 3 · Breadboard 4 · Prototype 5–6 · Validated 7 · Operational 8–9 (hardware analogue of PoC⟶MVP⟶α⟶β⟶release). Org rows = operational deliverables. ✓ = in place pre-award; Dome-Animals cells = per-species level. Each Progress Milestone needs a quantitative exit criterion (FOA §9.2). Stages illustrative.

Initials	Institution	Model
BS	CU Anschutz	Ferret
MD	UC Irvine	Guinea fowl
DF	CCMH	Mouse
AH	UCLA	Marmoset, NHP
MH	UT Austin	Human
KB	Indiana Univ.	Human, NHP
JY	UC Berkeley	Human, NHP
GN	RAI Institute	Robotics
CH	FAMU-FSU	Robotics
JH	Agility Robotics	Robotics
SH	Cornell	Robotics

Table 2: External research collaborators spanning every model system and dimensional domain the Dome instrument serves.

2. Technology Landscape

The perception–action loop is measured today by six mature but separately-evolved instrument ecosystems, each excellent within its own thread. Body kinematics are captured by markerless pose estimation (DeepLabCut, SLEAP, ViTPose, FreeMoCap), marker-based systems (Vicon, Qualisys, OptiTrack), and inertial suits (Xsens); the forces behind that motion by force plates (AMTI, Bertec), instrumented treadmills, and pressure-sensing insoles; gaze by wearable eye trackers (Pupil Labs, Tobii); neural activity by Neuropixels probes, miniscopes, and mobile

DRAFT - Work in Progress

EEG; muscle activity by surface and intramuscular EMG; and the surrounding scene by photogrammetry, Gaussian splatting, RGB-D, and LiDAR. Every one of these threads is precise and improving along its own trajectory. Progress within any single channel is not the bottleneck.

The bottleneck is that these instruments do not compose. Each defines its own coordinate frame, runs on its own clock, and names its quantities in its own scheme — so a "gaze point" in one system is not the same measured quantity as in another — and none of them carry the uncertainty of a measurement across a modality boundary. Assembling the whole loop therefore falls on each lab as bespoke, one-off integration. Mobile Brain/Body Imaging[1] is the fifteen-year demonstration of this ceiling: it is a methodology rather than an instrument, because the integration burden it describes is handed anew to every team that adopts it, and its results rarely leave the Methods sections of the labs that built them.

The deeper reason integration never sticks is that today's platforms are organized around sensors rather than around measurements. A tool is built for a particular camera model or force-plate vendor, not for kinematics as such — even though kinematics is kinematics whether it came from cameras or from IMUs — so records from different labs, species, and hardware generations cannot be pooled, compared, or cumulatively improved. The Dome inverts this. It is defined by the measurement, carries an unbroken chain of calibration and propagated uncertainty from raw sensor to reconstructed quantity, and therefore yields the same canonical, uncertainty-tagged record regardless of which instruments happen to be present. This is the capability no existing system provides: not a better sensor in any one thread, but a single instrument in which every thread is spatially calibrated, temporally synchronized, and semantically unified.

Two adjacent fields are stalled for want of exactly this record. Computer-vision pose estimation has drifted away from the pixels toward ungroundable targets — 3D-from-2D, mesh fitting, semantic segmentation — while still benchmarking against COCO imagery scraped from the early 2000s, and it fails on the out-of-distribution movement that matters most: clinical gait, extreme athletics, animal locomotion. It needs physical ground truth at scale, which a calibrated multi-camera Dome produces by construction. Legged-robot reinforcement learning is likewise limited not by simulation — MuJoCo, Isaac Lab, and Genesis are excellent — but by the sim-to-real gap, and the resource that would close it is uncertainty-tagged, multimodal recordings of real agents in real environments; the standard corpora (Human3.6M, AMASS, GRAB) are single-modality, lab-constrained, or carry no uncertainty at all. The same Dome record feeds both. Building the instrument that makes the loop itself the unit of measurement is the goal of this proposal.

3. Outcomes

3.1. The instrument

Our five-to-seven-year outcome is a single instrument — the DOME — realized in three human-scale forms and an animal-scale form that records the complete perception–action loop as one calibrated, uncertainty-tagged measurement and yields the same canonical record regardless of which sensors are present. DOME-L is the warehouse-scale flagship in Greater Boston; it drives each sub-instrument to its limit and is large enough to physically contain the smaller forms, which makes it the metrological reference for the entire network. DOME-S extends FreeMoCap's webcam capture into the disseminated lab- and classroom-scale form, and DOME-Mobile is the wearable form that carries the measurement onto natural terrain outside the lab. Because the instrument is defined by its measurement ontology rather than by any one sensor, a markerless-camera estimate and an inertial estimate describe the same kinematics and fuse into a single estimate better than either alone.

3.2. The organizing benchmark

The benchmark that organizes the entire build is veridical retinal input: reconstructing the world projected onto a moving agent's retina to within 1° at the fovea while it fixates a point on the ground at a 45° angle. This target is defensible rather than aspirational because it composes two independently measured error sources — joint-angle accuracy already characterized in FreeMoCap[2] and the published errors of commercial eye trackers — into one explicit uncertainty budget. Closing that budget is what forces the hardware advances below, so we state the number once and build toward it.

3.3. Body — trustworthy kinematics

On the motor half of the loop we deliver kinematics clean enough for inverse dynamics from markerless capture. The actuated, self-calibrating camera array places daisy-chained Power-over-Ethernet cameras on programmable mounts that stay metrically calibrated as one instrument while being re-aimed and re-focused from a single console, dissolving the bottleneck that leaves large-volume rigs frozen in place because manual re-calibration is too costly. This is adaptive AI-based sensing in hardware: the algorithm reconfigures the physical optics to maximize observability, running autonomous experiments over camera geometry rather than settling for a hand-placed rig. Success is measured as post-reconfiguration calibration quality and the time from sub-volume selection to calibrated capture, against a fixed-calibration reprojection baseline. Camera-to-IMU fusion then combines noisy-but-accurate outside-in estimates with precise-but-drifting inside-out estimates under explicit per-joint uncertainty, validated as joint-torque uncertainty against force-plate ground truth and muscle-force uncertainty against EMG-informed estimates.

3.4. Eye — the flagship transducer

On the sensory half we build the flagship transducer, a binocular eye tracker that measures degrees of freedom commercial trackers omit. It resolves ocular torsion from iris texture — the degree of freedom that corrupts the curl of retinal optic flow and a standing research target[3] — targets microsaccade-scale resolution, and pursues lens accommodation via dual-Purkinje imaging as a stretch goal. The design bet is a dense multimodal sensor array built like a phone camera bank that trades low profile for data quality, which makes it a camera-quality engineering bet rather than a physics one and keeps unit cost on the commodity-imager curve, so that next-generation eye tracking can saturate the field the way FreeMoCap did for motion capture. Its benchmarks are sub-15-arcminute eye-in-head accuracy, 1° torsion accuracy, and the composed 1° gaze-in-world target above.

3.5. World — the scene the agent acts in

Reconstructing what the eye sees requires reconstructing the world it acts in, so a head-mounted world-scanner of stereo-RGB, structured-IR, and LiDAR produces per-frame RGB-D and, over longer horizons, a terrain mesh and a SLAM-corrected head trajectory that fixes inertial drift[4]. The same scanner package mounts on drone-swarm nodes that act as a mobile extension of the camera array, grounding DOME-Mobile's inside-out estimates outdoors where a fixed rig cannot reach, benchmarked as gaze and kinematic uncertainty against the DOME-L reference during co-recorded trials. Modular floor panels — each a force plate, terrain tile, or LED surface — and the ARGP perturbation platform close the world half of the loop by making the ground itself both sensor and controllable stimulus.

3.6. One record across species

The measurement crosses species, which is the claim that this reshapes a field rather than sharpening a tool: functionally equivalent instruments at collaborator labs yield the same canonical records as the human instrument. The animal-scale instrument already exists — the team's ferret and mouse builds integrate three-camera gaze, full-body kinematics, and world cameras in one calibrated system, with neural recordings being added now — and the human, mouse, and ferret builds are being brought to functional equivalence, extending to guinea fowl for musculoskeletal biomechanics with EMG MD and marmoset for primate electrophysiology AH/JY. Success is cross-species measurement comparability under shared calibration and data-exchange protocols, so that a reconstructed `RetinalInput` or `RigidBody` denotes the same thing whether it came from a human, a ferret, or a marmoset.

3.7. What the record unlocks

Because every channel is calibrated, synchronized, and uncertainty-traced within one sensor-grounded ontology, the record yields capabilities that would each otherwise demand a bespoke project. The instrument generates its own training signal by back-projecting calibrated 3D reconstructions onto each camera's 2D view, and the resulting reprojection error fine-tunes pose estimators, measured as improvement on out-of-distribution movement — clinical gait, acrobatics, animal locomotion — rather than on a saturated benchmark: an instrument engineered for next-generation AI training pipelines. The record is natively reinforcement-learning-ready because a DOME export maps directly onto MuJoCo and Isaac Lab rigid-body models with uncertainty intact, so DOME-recorded corpora drive sim-to-real policy transfer and inverse-RL recovery of the control policies underlying observed behavior, testable back in DOME-L via ARGP perturbation.

3.8. Dissemination and validation

The outcome is a countable, validated footprint rather than a released codebase: [N] validated DOME-S instances spanning human biomechanics, robotics, visual neuroscience, and clinical sites, each cross-validated against the DOME-L reference and shipped with a build guide, parts list, and calibration protocol any lab can follow. We anchor clinical rigor to FDA 510(k) clearance of the FreeMoCap/DOME instrument for rehabilitation and assessment, letting a regulatory bar drive protocol quality. Phase activities and maturity targets for every line above appear in the Milestone Matrix (Table 1), whose quantitative exit criteria are the benchmarks named here.

4. Senior/Key Personnel

Under the NSF X-Labs model, Phase 0 is the runway for members to transition to full-time before Phase 1. The PI and CSO are already full-time on FreeMoCap; the others commit to full-time by Phase 1, most having already contributed for years — part-time or volunteer, alongside senior industry and academic roles — before any funding existed.

Jonathan Matthis, PhD — President / PI. Founded and maintains FreeMoCap; two decades measuring the full perception–action loop. Left a tenure-track professorship (Human Movement Neuroscience, Northeastern) because it could not support the integrated tool-building this work requires. NEI NIH K99/R00 recipient (1,400 citations), with a publication record spanning the full loop, from outdoor gaze-and-gait to real-world retinal-input reconstruction.

NR — Chief Executive Officer. Technology-sector engineering leader and startup co-founder, recruited to run the organization so the PI can focus on the technical Mission. Staff Platform Software Engineer at Rapid7; co-founded and scaled MailLift, a venture-backed API startup. Brings the leadership and operations capacity to build FreeMoCap into an autonomous, independent X-Lab.

EI — Chief Technology Officer. Enterprise software architect (15 years, engineer to Principal Architect / CTO) and FreeMoCap's CTO since 2021, essentially since founding. Concurrent CTO at Chorus Innovations; former Principal Architect at Unqork; distributed-systems and data-engineering depth (Python, Node.js, PostgreSQL, Spark; RDF/SPARQL). Ensures enterprise-grade architecture and the calibrated, semantically-unified data backbone the instrument requires.

AC, PhD — Chief Scientific Officer. Dissertation validated FreeMoCap against research-grade optical motion capture (Qualisys), demonstrating clinically valid kinematics from commodity hardware. Expertise in clinical tool validation, bench-to-bedside development, neuroprosthetics, and brain imaging; leads DOME-S development and clinical dissemination.

JKL, PhD — Chief AI Officer. Cognitive scientist and applied-AI architect (Applied AI Architect, SOLID Inc.); FreeMoCap contributor since 2024 and lead developer of SkellyBot, its multi-agent AI assistant (NestJS + LangChain, vector-store retrieval, multimodal inputs). 15+ years handling sensitive human-subjects and health data under FERPA and IRB — anchors data governance and research security, and builds internal AI for development, support, and education.

RR — Chief Financial Officer. Cooperative- and nonprofit-finance specialist and FreeMoCap's Treasurer / volunteer CFO since 2021, its longest-serving financial officer. Founder of Capital Bookkeeping Cooperative; Controller at Honest Weight Food Co-op; Adjunct Professor of Accounting, University at Albany (MS, Accounting). Runs financial operations and grant management; cooperative-governance experience fits the Foundation's open structure and the X-Labs autonomy requirements.

KM, PhD — Project Manager, DOME-Mobile. Computer-vision engineer, data scientist, and longtime PI co-author whose work maps almost one-to-one onto DOME-Mobile. Built the outdoor eye-tracking + photogrammetry + IMU pipeline the DOME generalizes[4,5]; currently Senior Data Scientist at SynMax (multi-sensor fusion, cloud MLOps). Holds essentially the complete skillset for DOME-Mobile's core instrumentation.

MN — Project Manager, DOME-L. Clinical biomechanics, medical-systems design, mechatronics, and multi-modal mocap-lab management; holds the skillset for large-scale DOME-L facility design and operation.

5. Team Capabilities Statement

Building the Dome demands a combination of expertise that no single institution's faculty could assemble, because the work is neither pure science nor pure engineering but the sustained craft of turning one into the other. The FreeMoCap Foundation was built from the ground up to hold that combination under one roof. The leadership team pairs scientific domain expertise in perception–action neuroscience and clinical biomechanics with enterprise software architecture, applied-AI systems, and hardware and mechatronics engineering at both wearable and facility scale, all run by dedicated operations and finance leadership recruited so the technical staff can stay on the Mission. Each member is detailed individually in § 4; what matters here is that they already function as a single team, having built and maintained a shipping instrument together — most of them for years, part-time or volunteer alongside senior industry and academic roles, before any funding existed. That is the adaptability and collaborative track record the Mission requires, demonstrated rather than promised.

5.1. Governance and autonomy

The FreeMoCap X-Lab (FMC-X) will operate as an independent project within the FreeMoCap Foundation, a 501(c)(3) nonprofit whose leadership is nearly identical to FMC-X's and whose existing mission almost fully overlaps it. Final decision authority rests with the President/PI at the top of the org chart, so research, partnership, and hiring decisions are made in days rather than weeks, with no higher management to consult. The Foundation is already autonomous as of this submission: it holds full internal control of funding allocation, research direction, partnership agreements, intellectual property, and hiring, requires no parent-institution approval for any operational decision, and satisfies every condition of the NSF X-Labs Autonomy Factor Test (§6.1.1) at the time of submission. Phase 0 is the runway over which the remaining leaders transition to full-time ahead of Phase 1; the PI and CSO already are.

5.2. Network, partnerships, and community

A standing, active network of research groups already spans every domain the Dome serves — human perception-and-action researchers, robotics and prosthetics groups, visual neuroscientists working in ferret and mouse, musculoskeletal biomechanists working in guinea fowl, and primate electrophysiologists. This network is real and in use today, not aspirational, with existing collaborations and shared tooling across multiple sites; the PI architects and holds the instrument-and-network together while the domain-specific science is executed by the named collaborators (Table 2). The community sets design and roadmap priorities through structured processes that gather broad input without stalling on consensus — a Request-for-Comments process modeled on Python's PEP system — and convenes in person twice a year across workshops that train students and out-of-domain experts, hackathons that onboard developers, conferences that share Dome -based research, and a congress that sets organizational direction. A Community Grants Program and Developer Fund sustain the open-source network and double as a testbed for engagement mechanisms.

5.3. A proven model at smaller scale

FreeMoCap has already proven this operating model. It is an open-source markerless motion-capture instrument, built and maintained outside academia, scoped to the measurement itself with an obsessive focus on usability and refusing to claim any research domain — and that refusal is precisely what let it become a commons rather than one lab's tool. The receipts are concrete: over 15,000 users across 152 countries, more than 10,000 GitHub stars, and over 3,500 Discord members. The Dome applies the same measurement-scoped, master-built, open, full-time-maintained model to the far larger target of the complete interaction loop — a scale reachable only with the dedicated, autonomous, milestone-based funding the X-Labs program exists to provide.

5.4. Intellectual property and dissemination

The IP strategy keeps the core instrumentation open while actively pursuing adoption and commercialization: an open-source core under permissive licensing for downstream use, structured so the platform technologies stay widely accessible as the team pursues adoption, commercialization, and ecosystem growth. A full IP Management Plan is due by the end of Phase 0 under the OT agreement and will be developed against that milestone.

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